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Real-Time Segmentation and Feature Extraction of Electromyography: Towards Control of a Prosthetic Hand

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Abstract: Surface electromyographic (sEMG) signals can be used as inputs to control a myoelectric prosthetic hand. This requires the discrimination of sEMG associated with different hand movements. The signals for key unilateral hand movements such as wrist extension, wrist flexion, and power grip are similar, making the classification and control of these hand movements challenging. This preliminary study explores the viability of classifying the sEMG signals of these hand movements in real-time, in order to control a software simulation of a prosthetic hand. sEMG data was recorded from two bipolar electrodes for offline classifier training purposes. A novel segmentation technique was used to separate the muscle contraction and rest periods of the sEMG time-series data. A time-frequency algorithm was then applied for the first time to extract sEMG-based features from the segmented data. Features were used to train four support vector machines offline, in a one-versus-all architecture. The classification system was tested offline and in real-time. The system yielded accuracies of 89.39% and 84.93% for offline and real-time testing respectively. Real-time classification was done in 10.2 ms when processing 0.5 s of input sEMG data. This shows that the system is accurate and computationally efficient, even with limited electrodes. Thus, it can serve as a foundation for further work in the implemented segmentation and feature extraction methods, for an inexpensive alternative to commercial myoelectric prosthetic devices.

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Keywords: sEMG, segmentation, feature extraction, SVM.

1. INTRODUCTION

Following the partial or complete amputation of a limb, people lose the ability to perform several activities of daily living (ADLs) (Sollerman & Ejeskär 1995). As a result, many amputees seek prostheses that are usually body-powered or myoelectric (Young et al. 2013). Myoelectric prostheses allow for more intuitive control than body-powered prostheses, as well as a greater range of degrees of freedom (Young et al. 2013). Intuitive control means that the prosthesis is neurally controlled in a manner similar to that of the lost limb. Myoelectric prostheses commonly use surface electromyography (sEMG) to measure the electrical activity of muscles as control inputs to the device (Shenoy et al. 2008). Classification techniques make it possible to identify intended movements from features extracted from the obtained sEMG signals (Young et al. 2013). Such prostheses allow amputees to perform some of the lost ADLs and in turn improve their quality of life (Shenoy et al. 2008).

To control a myoelectric prosthetic hand (MPH), sEMG information from an amputee's remaining muscles is required to be measured and processed. Segmentation is used to identify periods of contraction (corresponding to periods of movement) and periods of rest (occurring in between contractions). Features can be extracted from these periods for use in a learning algorithm, that can in turn be used to control a MPH. The aim of this research is to extract more relevant information from the sEMG signals to yield more accurate and intuitive control.

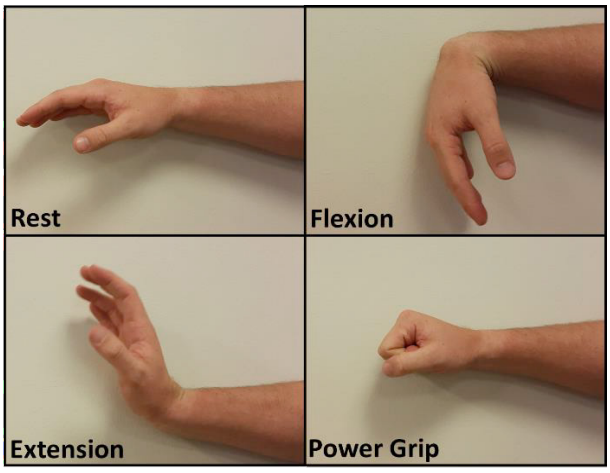

Figure 1: Static hand movements.

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Segmentation of sEMG signals can be performed manually or by correlating muscle contraction to a video recording of the movement (Sedlák et al. 2013). These methods are time consuming and are improved by automatic segmentation algorithms (El Falou et al. 2005). Owing to the non-stationary properties of sEMG (Seven et al. 2013), it is possible to use time-frequency analysis to characterise the changes in sEMG signals over time. An algorithm, previously applied to music recognition (Haitsma & Kalker 2003), is adapted and applied to sEMG, which to the authors' knowledge has not been done before. Other feature extraction methods include variance, the energy of wavelet coefficients of sEMG in nine scales and Cepstrum coefficients (Boostani & Moradi 2003).

In this study, four static hand movements are classified both offline and in real-time to assess the validity of the novel segmentation and feature extraction processes. These movements can be seen in Figure 1: rest, wrist flexion, wrist extension, and power grip are key to allow the performance of ADLs (Trombly & Radomski 2002). Support vector machine (SVM) classification is used to validate the segmentation and feature extraction processes (Shenoy et al. 2008). The latter is the focus of this paper, aimed at improving the discrimination of sEMG of key hand movements and is an initial step towards controlling a MPH (represented in this study by a simulated prosthetic hand) in real-time.

An overview of the system and data collection process is presented. This is followed by details of the novel segmentation method and the adapted sEMG feature extraction technique. The classification system and simulations used for testing are discussed. Thereafter, a discussion and future recommendations are presented.

2. METHOD

2.1 Overview

The system block diagram is shown in Figure 2. Each aspect of the system is further elaborated in the upcoming sections.

2.2 Data Collection

All recorded datasets (offline and real-time) were filtered by a 4th order bandpass Butterworth filter of bandwidth 20 – 450 Hz (Young et al. 2013; De Luca et al. 2010). Two offline sEMG datasets were recorded over the period of one day. Dataset 1 was used for training of the classification system and preliminary offline testing. Dataset 2 was used as a simulation of real-time processing.

Data was collected from two, right-handed, healthy, young male subjects (Subject 1 was age 24 and Subject 2 was age 23) over a total of 15 sessions each per dataset. sEMG data was recorded with a PowerLab 26T in conjunction with LabChart 7 (both by ADInstruments) and was processed in MATLAB. The two available channels for sEMG recording were used, where bipolar sEMG electrodes were placed over the right forearm flexor (channel 1) and the right forearm extensor (channel 2) muscle groups (Criswell et al. 2011). The ground electrode was placed on the left forearm.

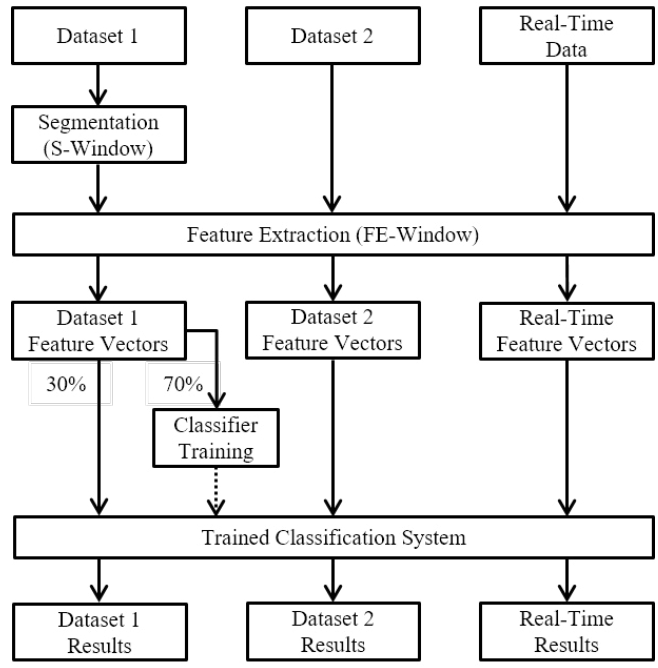

Figure 2: System block diagram.

The subjects were seated with the right arm resting on a level surface. Five sessions were performed for each movement i.e. wrist flexion, wrist extension and power grip. A session began at rest and was comprised of 20 repetitions of one of the three movements. Each repetition consisted of three seconds of sustained contraction followed by four seconds of rest. There were two minutes of rest between sessions. For the power grip sessions, contractions were performed at approximately 50% maximum voluntary contraction (MVC) while gripping a stick of diameter 2.7 cm (Lazaro et al. 2014). The data was sampled at 2 kHz (Shenoy et al. 2008) and was collected simultaneously for both channels. The data was filtered with a 50 Hz notch zero-phase-lag FIR filter by the LabChart software to remove line noise. Each subject was instructed not to observe the sEMG signals during recording as visual bias might have been introduced (Shenoy et al. 2008).

After development of the system, real-time sEMG data was recorded from the same two subjects under the same device settings and filtering as before. This was used for real-time testing. The data was passed to MATLAB using the real-time LabChart to MATLAB module. 1024 voltage-time values were passed to a buffer in MATLAB, which corresponded to approximately 0.5 s of data.

2.3 Segmentation

Segmentation was performed on each channel of recorded data per session from Dataset 1. Owing to the highly variable amplitudes that occur in sEMG signals, each voltage-time value in the channels was amplified per Equation 1, before segmentation could occur.

$$channel_{amp}(t) = |\, channel(t) \,|^{1.5} \qquad (1)$$

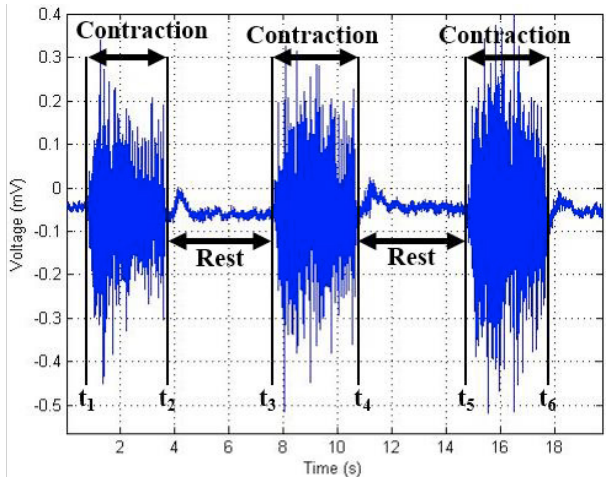

Figure 3: Raw sEMG data showing contraction and rest periods and corresponding times prior to segmentation.

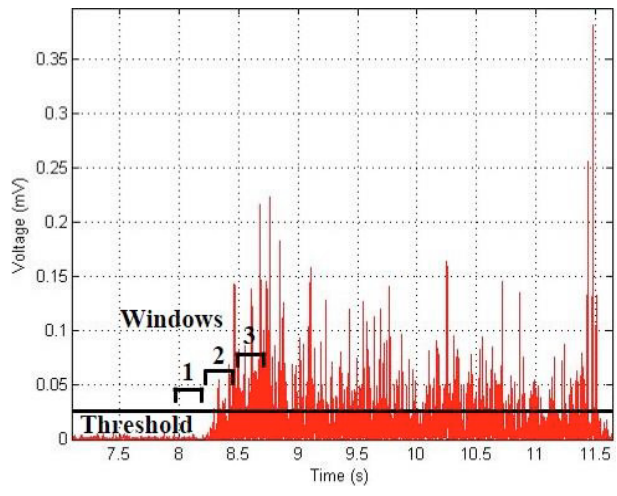

Figure 4: Segmentation procedure with a selection of non-overlapping S-Windows. Window 2 is the start of contraction.

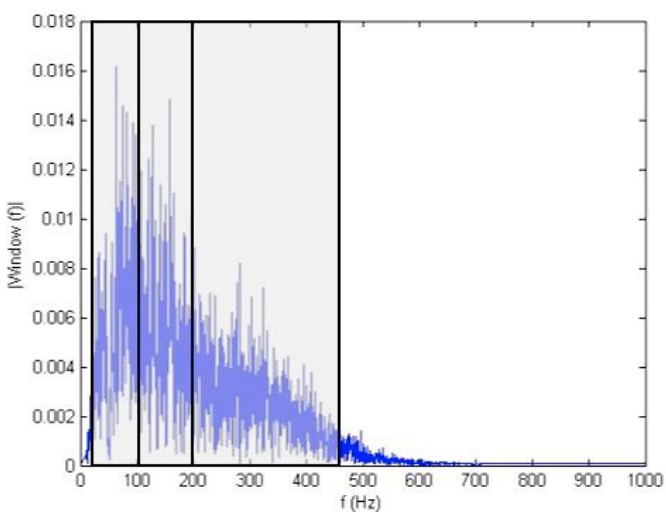

Figure 5: Three frequency bands for feature extraction: 20 – 100 Hz, 100 – 200 Hz and 200 – 450 Hz.

Table 1: Factor *n* and S-Window size for the flexor and extensor muscle groups in the right forearm.

Muscle Group	Factor *n*	S-Window Size
Flexor	15	300
Extensor	6	600

It is not suitable to differentiate contraction and rest periods (see Figure 3) based on amplitude or static threshold values alone, as the contraction periods may contain smaller values which mimic rest values. The initial recorded portions of each channel did not involve contraction and was thus used to determine a session-specific threshold according to Equation 2.

$$threshold = n\ (mean\ (rest) + 2\sigma_{rest}) \tag{2}$$

n is a factor that is utilised depending on the muscle group from which the channel data was collected (see Table 1), σ is the standard deviation, and *rest* denotes the first known rest period in the sEMG recording. The use of 'mean plus twice standard deviation' is used to determine most values within a range of data. Assuming a normal distribution, this encompasses 97.5% of values and should exclude the high value outliers in the range. For example, during a rest period a muscle twitch may occur which introduces high voltage values which are therefore ignored.

The amplified sEMG time-series data, $channel_{amp}(t)$, was then windowed using a segmentation window (S-Window) from the start of the recording (see Figure 4), in order to distinguish rest from contraction periods using Equation 3.

$$mean\ (window) + 2\sigma_{window} > threshold \tag{3}$$

window refers to the time-series data within an S-Window. The size of the applied S-Window is dependent on the muscle group being analysed (see Table 1) and the S-Window is a non-overlapping, rectangular window with unity amplitude. The left-hand side of Equation 3 was determined from the current S-Window. When the threshold condition was met, it corresponded to the start of a contraction period and thereafter, when the threshold condition was no longer met, it implied the end of a contraction period. Once these times were known (see Figure 3), they were used to extract segments of rest and contraction from the raw sEMG data in Dataset 1. These segments were saved by session to label each segment.

Equations 2 and 3 were derived by observation of the experimental data. The values of the factor *n* and S-Window size were determined by experimentation. These values were dependent on the muscle group being actively used in the recording (regardless of the movement being performed).

2.4 Feature Extraction

The segmented data was windowed by a feature extraction window (FE-Window). The FE-Window is a sliding rectangular window of unity gain and length of 512 samples (approximately 0.25 s). The FE-Window was shifted by 256 samples on each application, creating a 50% overlap. A fast

Fourier transform (FFT) of each FE-Window was computed and the amplitudes of the data within three frequency bands were summed (see Figure 5) (Haitsma & Kalker 2003). Each sum formed a feature. These bands were determined by observation of the FFTs of the FE-Windows for contraction segments and assessing where most energy was contained.

Each FE-Window resulted in three features per channel. This yielded a feature vector containing six features for each FE-Window in time (from channel 1 and channel 2).

For each subject, 1425 feature vectors were extracted in total and labelled for each movement's segments and for rest segments. This was used for training and testing purposes.

2.5 Classifier Training, Classification and Display

SVMs are binary classifiers used in supervised classification problems and can be used in sEMG classification (Shenoy et al. 2008). Although SVMs are binary, they are commonly used where more than two classes are required, given their robustness and ability to obtain high classification accuracies (Shenoy et al. 2008). To achieve this, multiple binary SVMs are combined in either a one-vs-one (OVO) or a one-vs-all (OVA) architecture (Kim et al. 2010). The OVA architecture requires the same number of SVMs as there are classes, while the OVO algorithm requires $k(k\text{-}1)/2$ SVMs, where k is the number of classes (Kim et al. 2010). Both architectures have low computational requirements (Oskoei & Hu 2008).

Offline Classifier Training: Each hand movement corresponds to an individual class, so for the OVA algorithm, four SVMs were trained. Together, these four SVMs and the OVA algorithm are referred to as the classification system (see Figure 6A). Each SVM's binary classes are defined as a relevant movement and a non-relevant movement; for example, flexion and non-flexion. Feature vectors extracted from Dataset 1 were used to train each SVM, where 70% of the feature vectors were used for the training, and the remaining 30% were used for offline testing (Govada et al. 2015), as discussed in Section 3.1.

Real-Time Classification: For every 0.5 s of recorded data (a buffer of sEMG data), three feature vectors were extracted using FE-Windows and input to the classification system (see Figure 6A). Each SVM classified the three inputs independently, and a single output for the SVM was determined by a majority vote (see Figure 6B).

The OVA algorithm was implemented in the decision comparison sub-system (see Figure 6A). It compared the outputs of the four SVMs to determine a final classification system output. For a movement to be simulated, the SVM responsible for that movement must output 'true', and the other SVMs must output 'false', otherwise the system will output an unknown hand movement.

Simulation Display: The classified hand movement was displayed on a screen, so that a user had visual feedback while the system was active. Rest was the default hand movement. If the classification system output was an 'unknown hand movement', the classification system outputted a notification and returned to its default state. This ensured that the simulation display returned to a known state, so that subsequent hand movements were displayed correctly.

2.6 Testing Procedures

The system was tested using three metrics for both subjects. These were offline classification, real-time classification, and robustness. The results of these tests can be seen in Table 2.

Offline classification assesses the quality of the chosen features for the system, compared to other feature extraction methods (Boostani & Moradi, 2003). There are two aspects to offline testing – cross-validation was performed using Dataset 1 and simulated real-time testing was done using Dataset 2. The classification scores of these two tests were averaged, yielding an overall offline classification result.

Real-time tests were performed for each movement. The test began at rest, and after each movement, the subject returned to rest. This was one repetition. In total, 20 repetitions of each movement were performed. The accuracy of the classification system was judged on its output from rest to steady state contraction and from steady state contraction to rest.

Robustness was measured by recording the number of misclassifications for a range of movements, excluding those explored in this study. These movements were performed at the shoulder, elbow, and wrist joints. The shoulder and elbow joint movements were not expected to result in a false positive output for the simulation, as the muscles responsible for these movements were not measured. The wrist joint movements were expected to result in some false positives, as the generated sEMG signals were similar to those in the study.

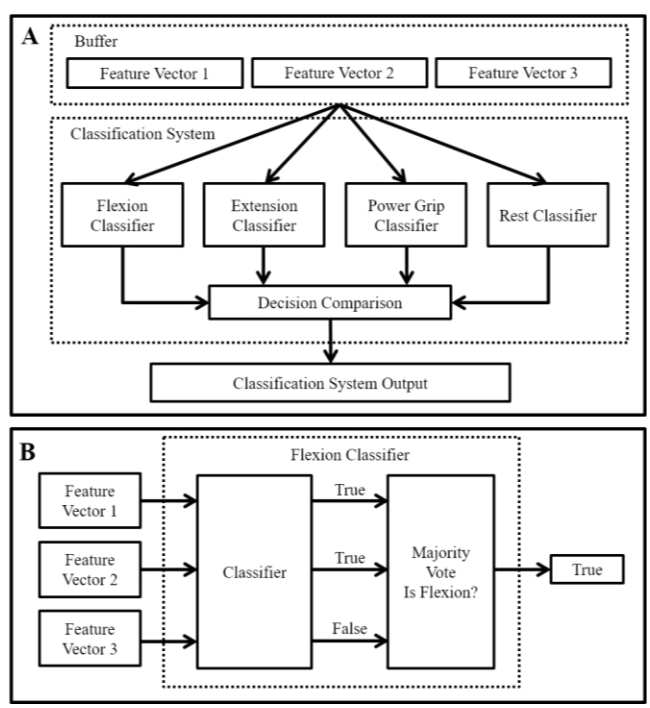

Figure 6: Classification system overview. **(A)** Classification system. **(B)** Flexion SVM decision making.

Table 2: Classification results.

Test	Subject 1 Accuracy (%)	Subject 2 Accuracy (%)
Offline Classification	90.49	88.29
Real-Time Classification	89.16	80.69
Robustness Misclassification	38.46	12.82

3. RESULTS

3.1 Classification Results

The system results for the tests described in Section 2.6 are shown in Table 2 for both test subjects. The average offline classification accuracy was 89.39% and the average real-time classification accuracy was 84.93%. The real-time classification test serves as the main classification success measure of the system, since the data was being processed in real-time, which is expected of an MPH. While promising, the classification accuracy should be higher before considering the use of this system in an MPH. The classification accuracy was improved through visual feedback (see Section 4.3).

Subject 1 displayed a high robustness misclassification rate. This may be owing to Subject 1 subconsciously flexing the right forearm during movements of the elbow joint. Further work will be required to explain this in more detail.

3.2 Performance

The system performance is quantified for feature extraction and classification. The feature extraction computation time was 5.9 ms on average and the classification computation time was 4.3 ms on average. These times were derived from real-time classification for 0.5 s of measured data as detailed in Section 4.3. This is sufficient for real-time hand movement control in this preliminary study. The system was tested on a Windows 8.1 laptop, with 8 GB RAM and an Intel Core i7 4702MQ 2.2 GHz processor.

4. DISCUSSION

4.1 Segmentation

The implemented segmentation method allowed for faster and less repetitive segmentation than the manual equivalent, and did not require video monitoring. It can result in false positives, as irregularities during the rest periods – such as muscle twitches – may occur. The amplitudes of these irregularities are relatively large compared to the rest period. In such cases, the manual segmentation method was employed. The algorithm may be improved by investigating the personalised factor n values further (see Equation 2).

S-Windows, shown in Figure 4, do not overlap, which may decrease the quality of the segmentation algorithm, owing to the lack of continuity in the windows. Consequently, transitions from contraction to rest (at the end of muscle contractions) were included in some segments but not others. These two factors resulted in few misclassifications and were therefore not of major concern, but if improved may yield a noticeable difference in classification accuracy. A different choice of S-Window type may also improve segmentation accuracy, which needs to be explored in further testing. The high classification rate suggests that the segmentation algorithm works automatically, and accurately in real-time.

4.2 Feature Extraction

The analysis of the sEMG data in the frequency range identified three generally consistent energy bands that may be used for effective feature extraction. The feature extraction method takes an average time of 5.9 ms to complete, which is computationally efficient and is suitable for real-time classification. The high classification rate suggests that the feature extraction method significantly differentiates between the movements investigated for each subject.

The average offline classification error of 10.61% is comparable to the offline SVM classification results of just over 5% error by Scheme & Englehart (2014). It is noted that the comparison is more indicative than diretly quantatative given that Scheme & Englehart (2014) used four electrodes as opposed to two and seven classes of movements as opposed to four (which are a subset). Multiple features including Cepstral coefficients and a number of time domain features were used to achieve their result (Scheme & Englehart, 2014).

4.3 Experimental Design

Possible inconsistencies in the experimental design included: a) during the collection of power grip data, 50% MVC was not measured, but rather approximated by the subjects; b) the offline classification tests were done while seated, whereas the real-time tests were done while sitting, standing with both arms at one's sides, and while standing with the right forearm flexed to 90°. In future, these tests should both be performed under the same experimental and environmental conditions.

Variation in the subjects' movement was kept to a minimum during the data collection process to reduce the noise in the acquired sEMG signals. The inclusion of noisy and varying data – while reducing the cross-validation accuracy – may be better for the system's robustness. Real-time training of the classification system will likely reduce the reliance on the data collection process (Shenoy et al. 2008).

It was observed that if visual or audio feedback was (even unintentionally) provided to the subject during the real-time classification testing, the subject could adapt their subsequent hand movements to ensure that they noticeably improved the system's identification of their movements (Shenoy et al. 2008). This indicates the importance of feedback in using such a system, since even if a classification system is trained well enough offline, the user is still able to adapt to improve the performance above and beyond what the system is believed to be capable of.

5. FUTURE RECOMMENDATIONS

Presently the system accounts for a total of four hand movements, including rest. Two additional movements, abduction and adduction of the wrist, were superficially

compared with the four movements in terms of their sEMG characteristics. It appeared likely that adding these movements in a six-class classification paradigm might be successful using the current system configuration. This will increase the number of movements that can be performed by an MPH.

Both S-Windows and FE-Windows were rectangular windows with unity amplitude. Different window types should be investigated, with varying overlap between windows. Furthermore, to validate the use of the feature extraction method, other classifiers should be implemented to identify more suitable classification algorithms.

0.5 s of buffered data was used in the classification system during real-time testing. This is too slow for a practical MPH as the response time of such a device should not be longer than 0.3 s (Englehart & Hudgins, 2003). Smaller buffer sizes should be explored while maintaining classification accuracy.

As shown by Young et al. (2013) and Shenoy et al. (2008), there is an inverse relationship between the number of electrodes and the misclassification rate. This leads to two potential further areas of work: maintaining the same number of electrodes or adding further electrodes, while classifying more movements. In an attempt to keep the cost of such a system to a minimum while still using two channels, the loss of accuracy when further movements are added, should be quantified. However, to adequately classify these movements, more channels may need to be utilised.

Each subject could achieve higher classification percentages by observing the simulation outputs (Shenoy, et al., 2008). This should be explored with further testing. Furthermore, the sample size of two subjects is not adequate to confirm the validity of the segmentation and feature extraction methods for use in an MPH. While the results of this preliminary study are promising, they should be confirmed by using more test subjects.

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